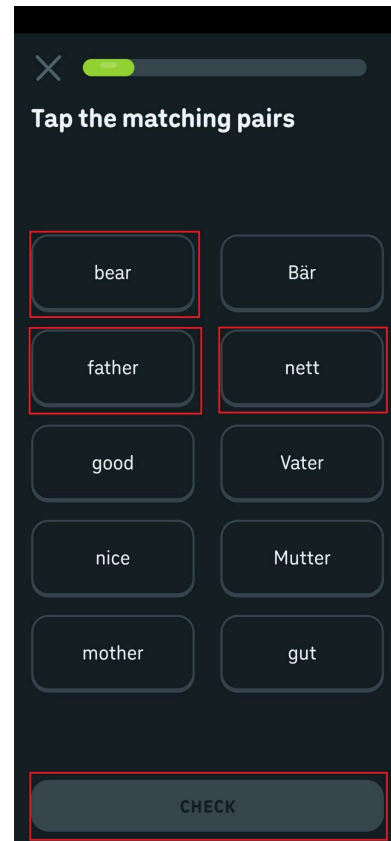
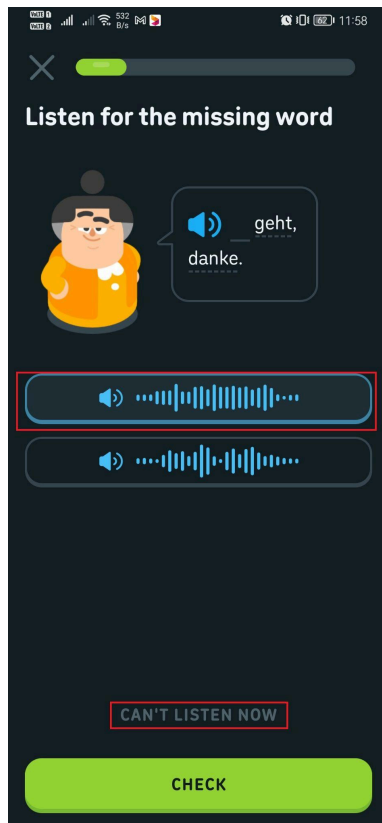


HCI Inclass Activity -36999

Application Name: Duolingo

1. Physical Differences

Design Implication: Catering to motor skill variations and sensory impairments.

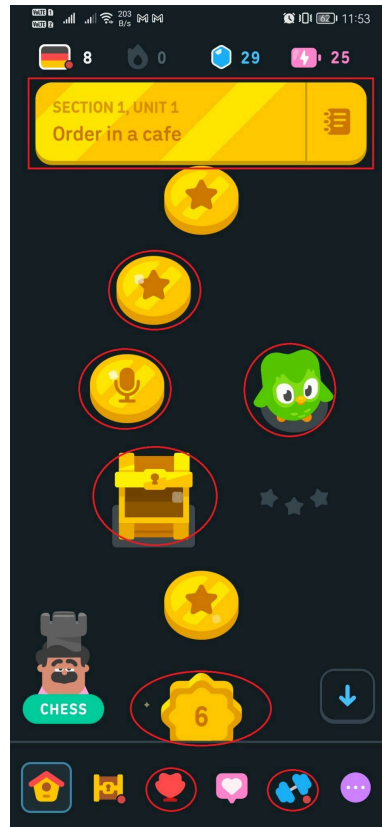


- **Large Touch Targets:** Duolingo uses oversized "word bubbles" and buttons. This caters to users with fine motor impairments or hand tremors, making it easier to select answers without precision errors.
- **Flexibility of Input:** The app includes "I can't speak/listen now" toggles. This is a crucial design implication for users with temporary or permanent physical disabilities (such as vocal or hearing impairments).

HCI Inclass Activity -36999

2. Effect of Age

Design Implication: Adapting to changing visual and attention spans across generations.

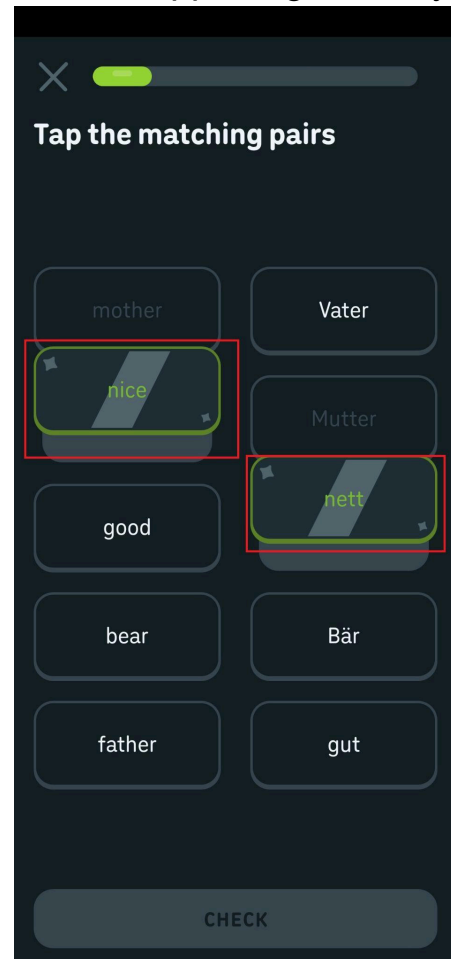
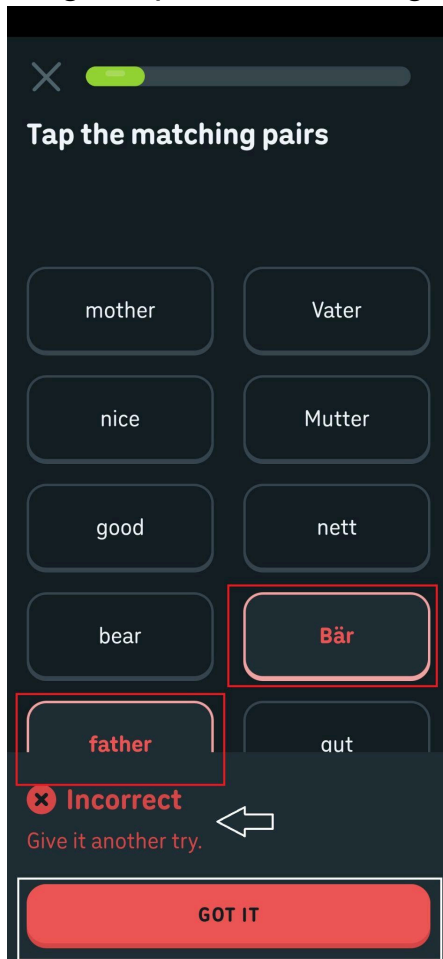


- **High Contrast & Legibility:** The interface uses a bold, custom sans-serif font and high-contrast colors. This caters to older users who may experience age-related vision decline (presbyopia).
- **Minimalist Visual Hierarchy:** By presenting only one primary task per screen, the app prevents older adults with declining selective attention from becoming overwhelmed by "visual noise".
- **Gamified Engagement:** For younger users, the "Streak" system and mascot (Duo) maintain motivation, while the simple, repeatable structure provides a comfortable routine for senior learners.

HCI Inclass Activity -36999

3. Cognitive Abilities

Design Implication: Managing mental workload and supporting memory.



- **Information Chunking:** Duolingo breaks complex language concepts into tiny, bite-sized lessons. This reduces Cognitive Load, ensuring the user's working memory is not overwhelmed.
- **Immediate Feedback Loop:** The app provides instant green/red feedback bars after every question. This caters to different cognitive processing speeds by allowing users to learn from errors immediately rather than trying to recall them later.
- **Recognition over Recall:** By providing word banks to choose from instead of a blank text box, the app reduces the "mental search" required, supporting users with varying levels of cognitive recall abilities.